

Major & Minor Scale Note Atlas

Sixty named major and minor references from PianoGrid's current chart: 15 major-key spellings and natural, harmonic and classical melodic minor for 15 tonics. Enharmonic names can share piano keys.
This is not a list of every scale or every fingering.

Generator: 2026-09-14-SCALES-COMPLETION-1. Current music input subset SHA-256 is recorded in the accompanying generation metadata.
No third-party source PDF, image, engraving or font file is embedded.

C Major

Ascending · low to high

Actual ordered pitches

Notes: C4 - D4 - E4 - F4 - G4 - A4 - B4 - C5

Fingers: Not source-checked; notes only

TREBLE CLEF

C4 D4 E4 F4 G4 A4 B4 C5

C4	D4	E4	F4	G4	A4	B4	C5
----	----	----	----	----	----	----	----

Descending · high to low

Actual ordered pitches

Notes: C5 - B4 - A4 - G4 - F4 - E4 - D4 - C4

Fingers: Not source-checked; notes only

TREBLE CLEF

C5 B4 A4 G4 F4 E4 D4 C4

C4	D4	E4	F4	G4	A4	B4	C5
----	----	----	----	----	----	----	----

Fingering coverage varies by selection; this atlas is a note reference.

Dh Major

Ascending · low to high

Actual ordered pitches

Notes: Db4 - Eb4 - F4 - Gb4 - Ab4 - Bb4 - C5 - Db5

Fingers: Not source-checked; notes only

TREBLE CLEF

Db4 Eb4 F4 Gb4 Ab4 Bb4 C5 Db5

Db4	Eb4	F4	Gb4	Ab4	Bb4	C5	Db5
-----	-----	----	-----	-----	-----	----	-----

Descending · high to low

Actual ordered pitches

Notes: Db5 - C5 - Bb4 - Ab4 - Gb4 - F4 - Eb4 - Db4

Fingers: Not source-checked; notes only

TREBLE CLEF

Db5 C5 Bb4 Ab4 Gb4 F4 Eb4 Db4

Db4	Eb4	F4	Gb4	Ab4	Bb4	C5	Db5
-----	-----	----	-----	-----	-----	----	-----

Fingering coverage varies by selection; this atlas is a note reference.

D Major

Ascending · low to high

Actual ordered pitches

Notes: D4 - E4 - F#4 - G4 - A4 - B4 - C#5 - D5

Fingers: Not source-checked; notes only

TREBLE CLEF



D4E4F#4G4A4B4C#5D5

D4	E4	F#4	G4	A4	B4	C#5	D5
----	----	-----	----	----	----	-----	----

Descending · high to low

Actual ordered pitches

Notes: D5 - C#5 - B4 - A4 - G4 - F#4 - E4 - D4

Fingers: Not source-checked; notes only

TREBLE CLEF



D5C#5B4A4G4F#4E4D4

D4	E4	F#4	G4	A4	B4	C#5	D5
----	----	-----	----	----	----	-----	----

Fingering coverage varies by selection; this atlas is a note reference.

F# Major

Ascending · low to high

Actual ordered pitches

Notes: Eb4 - F4 - G4 - Ab4 - Bb4 - C5 - D5 - Eb5

Fingers: Not source-checked; notes only

TREBLE CLEF

Diagram illustrating the ascending scale (low to high) for F# Major. The notes are Eb4, F4, G4, Ab4, Bb4, C5, D5, and Eb5. The diagram shows the notes on a staff and a corresponding piano roll visualization below it.

Eb4		F4		G4		Ab4		Bb4		C5		D5		Eb5
-----	--	----	--	----	--	-----	--	-----	--	----	--	----	--	-----

Descending · high to low

Actual ordered pitches

Notes: Eb5 - D5 - C5 - Bb4 - Ab4 - G4 - F4 - Eb4

Fingers: Not source-checked; notes only

TREBLE CLEF

Diagram illustrating the descending scale (high to low) for F# Major. The notes are Eb5, D5, C5, Bb4, Ab4, G4, F4, and Eb4. The diagram shows the notes on a staff and a corresponding piano roll visualization below it.

Eb4		F4		G4		Ab4		Bb4		C5		D5		Eb5
-----	--	----	--	----	--	-----	--	-----	--	----	--	----	--	-----

Fingering coverage varies by selection; this atlas is a note reference.

F Major

Ascending · low to high

Actual ordered pitches

Notes: E4 - F#4 - G#4 - A4 - B4 - C#5 - D#5 - E5

Fingers: Not source-checked; notes only

TREBLE CLEF

E4 F#4 G#4 A4 B4 C#5 D#5 E5

E4	F#4	G#4	A4	B4	C#5	D#5	E5
----	-----	-----	----	----	-----	-----	----

Descending · high to low

Actual ordered pitches

Notes: E5 - D#5 - C#5 - B4 - A4 - G#4 - F#4 - E4

Fingers: Not source-checked; notes only

TREBLE CLEF

E5 D#5 C#5 B4 A4 G#4 F#4 E4

E4	F#4	G#4	A4	B4	C#5	D#5	E5
----	-----	-----	----	----	-----	-----	----

Fingering coverage varies by selection; this atlas is a note reference.

F Major

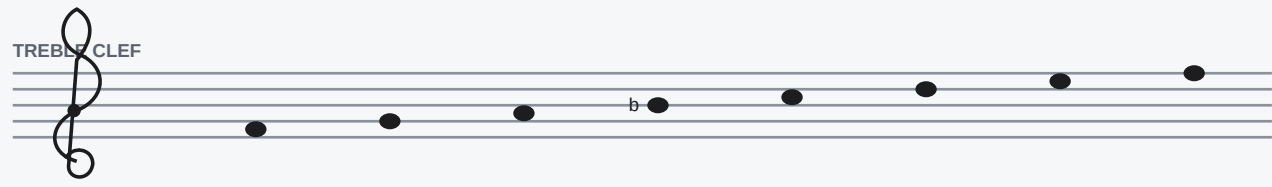
Ascending · low to high

Actual ordered pitches

Notes: F4 - G4 - A4 - Bb4 - C5 - D5 - E5 - F5

Fingers: Not source-checked; notes only

TREBLE CLEF



F4G4A4Bb4C5D5E5F5

F4	G4	A4	Bb4	C5	D5	E5	F5
----	----	----	-----	----	----	----	----

Descending · high to low

Actual ordered pitches

Notes: F5 - E5 - D5 - C5 - Bb4 - A4 - G4 - F4

Fingers: Not source-checked; notes only

TREBLE CLEF



F5E5D5C5Bb4A4G4F4

F4	G4	A4	Bb4	C5	D5	E5	F5
----	----	----	-----	----	----	----	----

Fingering coverage varies by selection; this atlas is a note reference.

F# Major

Ascending · low to high

Actual ordered pitches

Notes: F#4 - G#4 - A#4 - B4 - C#5 - D#5 - E#5 - F#5

Fingers: Not source-checked; notes only

TREBLE CLEF

F#4 G#4 A#4 B4 C#5 D#5 E#5 F#5

F#4	G#4	A#4	B4	C#5	D#5	E#5	F#5
-----	-----	-----	----	-----	-----	-----	-----

Descending · high to low

Actual ordered pitches

Notes: F#5 - E#5 - D#5 - C#5 - B4 - A#4 - G#4 - F#4

Fingers: Not source-checked; notes only

TREBLE CLEF

F#5 E#5 D#5 C#5 B4 A#4 G#4 F#4

F#4	G#4	A#4	B4	C#5	D#5	E#5	F#5
-----	-----	-----	----	-----	-----	-----	-----

Fingering coverage varies by selection; this atlas is a note reference.

G Major

Ascending · low to high

Actual ordered pitches

Notes: G4 - A4 - B4 - C5 - D5 - E5 - F#5 - G5

Fingers: Not source-checked; notes only

TREBLE CLEF

G4 A4 B4 C5 D5 E5 F#5 G5

G4	A4	B4	C5	D5	E5	F#5	G5
----	----	----	----	----	----	-----	----

Descending · high to low

Actual ordered pitches

Notes: G5 - F#5 - E5 - D5 - C5 - B4 - A4 - G4

Fingers: Not source-checked; notes only

TREBLE CLEF

G5 F#5 E5 D5 C5 B4 A4 G4

G4	A4	B4	C5	D5	E5	F#5	G5
----	----	----	----	----	----	-----	----

Fingering coverage varies by selection; this atlas is a note reference.

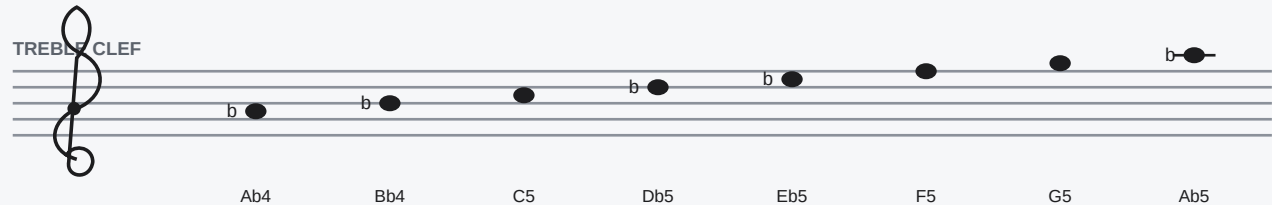
Ah Major

Ascending · low to high

Actual ordered pitches

Notes: Ab4 - Bb4 - C5 - Db5 - Eb5 - F5 - G5 - Ab5

Fingers: Not source-checked; notes only

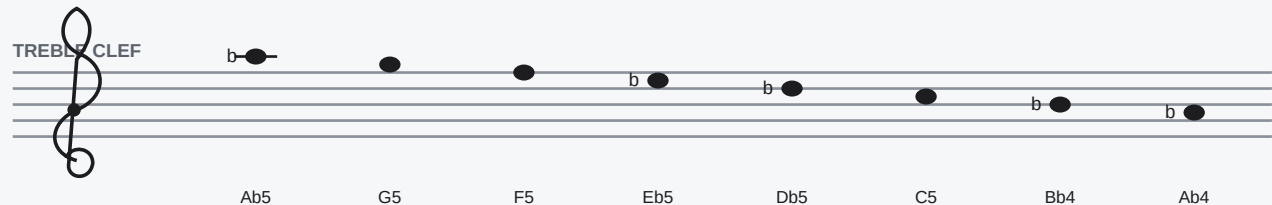


Descending · high to low

Actual ordered pitches

Notes: Ab5 - G5 - F5 - Eb5 - Db5 - C5 - Bb4 - Ab4

Fingers: Not source-checked; notes only



Fingering coverage varies by selection; this atlas is a note reference.

A Major

Ascending · low to high

Actual ordered pitches

Notes: A4 - B4 - C#5 - D5 - E5 - F#5 - G#5 - A5

Fingers: Not source-checked; notes only

TREBLE CLEF

A4 B4 C#5 D5 E5 F#5 G#5 A5

Descending · high to low

Actual ordered pitches

Notes: A5 - G#5 - F#5 - E5 - D5 - C#5 - B4 - A4

Fingers: Not source-checked; notes only

TREBLE CLEF

A5 G#5 F#5 E5 D5 C#5 B4 A4

Fingering coverage varies by selection; this atlas is a note reference.

Bb Major

Ascending · low to high

Actual ordered pitches

Notes: Bb4 - C5 - D5 - Eb5 - F5 - G5 - A5 - Bb5

Fingers: Not source-checked; notes only

TREBLE CLEF

Bb4 C5 D5 Eb5 F5 G5 A5 Bb5

Bb4								
	C5	D5		F5	G5	A5		Bb5

Descending · high to low

Actual ordered pitches

Notes: Bb5 - A5 - G5 - F5 - Eb5 - D5 - C5 - Bb4

Fingers: Not source-checked; notes only

TREBLE CLEF

Bb5 A5 G5 F5 Eb5 D5 C5 Bb4

Bb4								
	C5	D5		F5	G5	A5		Bb5

Fingering coverage varies by selection; this atlas is a note reference.

B Major

Ascending · low to high

Actual ordered pitches

Notes: B4 - C#5 - D#5 - E5 - F#5 - G#5 - A#5 - B5

Fingers: Not source-checked; notes only

TREBLE CLEF

B4 C#5 D#5 E5 F#5 G#5 A#5 B5

B4	C#5	D#5	E5	F#5	G#5	A#5	B5
----	-----	-----	----	-----	-----	-----	----

Descending · high to low

Actual ordered pitches

Notes: B5 - A#5 - G#5 - F#5 - E5 - D#5 - C#5 - B4

Fingers: Not source-checked; notes only

TREBLE CLEF

B5 A#5 G#5 F#5 E5 D#5 C#5 B4

B4	C#5	D#5	E5	F#5	G#5	A#5	B5
----	-----	-----	----	-----	-----	-----	----

Fingering coverage varies by selection; this atlas is a note reference.

Cb Major

Ascending · low to high

Actual ordered pitches

Notes: Cb4 - Db4 - Eb4 - Fb4 - Gb4 - Ab4 - Bb4 - Cb5

Fingers: Not source-checked; notes only

TREBLE CLEF

Cb4 Db4 Eb4 Fb4 Gb4 Ab4 Bb4 Cb5

Cb4	Db4	Eb4	Fb4	Gb4	Ab4	Bb4	Cb5
-----	-----	-----	-----	-----	-----	-----	-----

Descending · high to low

Actual ordered pitches

Notes: Cb5 - Bb4 - Ab4 - Gb4 - Fb4 - Eb4 - Db4 - Cb4

Fingers: Not source-checked; notes only

TREBLE CLEF

Cb5 Bb4 Ab4 Gb4 Fb4 Eb4 Db4 Cb4

Cb4	Db4	Eb4	Fb4	Gb4	Ab4	Bb4	Cb5
-----	-----	-----	-----	-----	-----	-----	-----

Fingering coverage varies by selection; this atlas is a note reference.

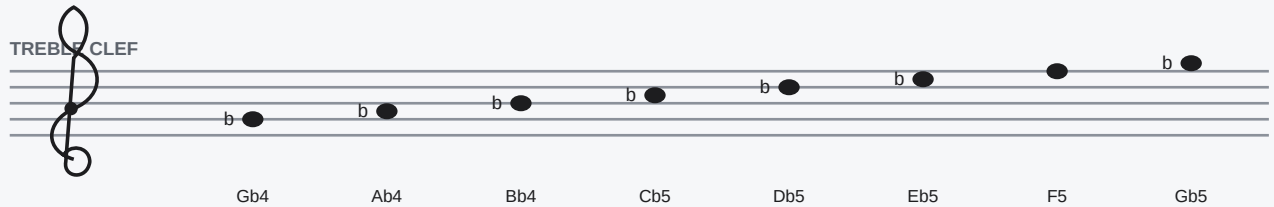
Gb Major

Ascending · low to high

Actual ordered pitches

Notes: Gb4 - Ab4 - Bb4 - Cb5 - Db5 - Eb5 - F5 - Gb5

Fingers: Not source-checked; notes only

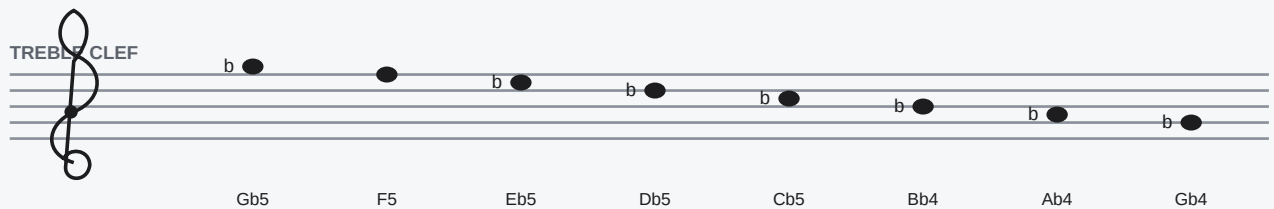


Descending · high to low

Actual ordered pitches

Notes: Gb5 - F5 - Eb5 - Db5 - Cb5 - Bb4 - Ab4 - Gb4

Fingers: Not source-checked; notes only



Fingering coverage varies by selection; this atlas is a note reference.

C# Major

Ascending · low to high

Actual ordered pitches

Notes: C#4 - D#4 - E#4 - F#4 - G#4 - A#4 - B#4 - C#5

Fingers: Not source-checked; notes only

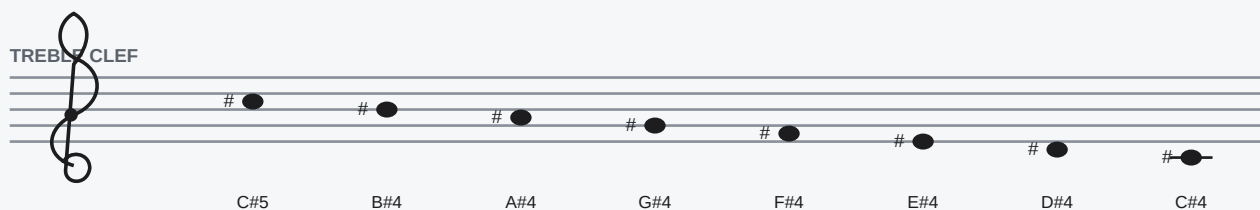


Descending · high to low

Actual ordered pitches

Notes: C#5 - B#4 - A#4 - G#4 - F#4 - E#4 - D#4 - C#4

Fingers: Not source-checked; notes only



Fingering coverage varies by selection; this atlas is a note reference.

A Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: A4 - B4 - C5 - D5 - E5 - F5 - G5 - A5

Fingers: Not source-checked; notes only

TREBLE CLEF

A4 B4 C5 D5 E5 F5 G5 A5

A4	B4	C5	D5	E5	F5	G5	A5
----	----	----	----	----	----	----	----

Descending · high to low

Actual ordered pitches

Notes: A5 - G5 - F5 - E5 - D5 - C5 - B4 - A4

Fingers: Not source-checked; notes only

TREBLE CLEF

A5 G5 F5 E5 D5 C5 B4 A4

A4	B4	C5	D5	E5	F5	G5	A5
----	----	----	----	----	----	----	----

Fingering coverage varies by selection; this atlas is a note reference.

F Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: E4 - F#4 - G4 - A4 - B4 - C5 - D5 - E5

Fingers: Not source-checked; notes only

Descending · high to low

Actual ordered pitches

Notes: E5 - D5 - C5 - B4 - A4 - G4 - F#4 - E4

Fingers: Not source-checked; notes only

TREBLE CLEF

E5 D5 C5 B4 A4 G4 F#4 E4

E4 F#4 G4 A4 B4 C5 D5 E5

Fingering coverage varies by selection; this atlas is a note reference.

B Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: B4 - C#5 - D5 - E5 - F#5 - G5 - A5 - B5

Fingers: Not source-checked; notes only

TREBLE CLEF

B4 C#5 D5 E5 F#5 G5 A5 B5

B4	C#5	D5	E5	F#5	G5	A5	B5
----	-----	----	----	-----	----	----	----

Descending · high to low

Actual ordered pitches

Notes: B5 - A5 - G5 - F#5 - E5 - D5 - C#5 - B4

Fingers: Not source-checked; notes only

TREBLE CLEF

B5 A5 G5 F#5 E5 D5 C#5 B4

B4	C#5	D5	E5	F#5	G5	A5	B5
----	-----	----	----	-----	----	----	----

Fingering coverage varies by selection; this atlas is a note reference.

F# Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: F#4 - G#4 - A4 - B4 - C#5 - D5 - E5 - F#5

Fingers: Not source-checked; notes only

TREBLE CLEF

F#4G#4A4B4C#5D5E5F#5

F#4	G#4			C#5			F#5
		A4	B4		D5	E5	

Descending · high to low

Actual ordered pitches

Notes: F#5 - E5 - D5 - C#5 - B4 - A4 - G#4 - F#4

Fingers: Not source-checked; notes only

TREBLE CLEF

F#5E5D5C#5B4A4G#4F#4

F#4	G#4			C#5			F#5
		A4	B4		D5	E5	

Fingering coverage varies by selection; this atlas is a note reference.

C# Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: C#4 - D#4 - E4 - F#4 - G#4 - A4 - B4 - C#5

Fingers: Not source-checked; notes only

TREBLE CLEF

C#4 D#4 E4 F#4 G#4 A4 B4 C#5

C#4 D#4 E4 F#4 G#4 A4 B4 C#5

Descending · high to low

Actual ordered pitches

Notes: C#5 - B4 - A4 - G#4 - F#4 - E4 - D#4 - C#4

Fingers: Not source-checked; notes only

TREBLE CLEF

C#5 B4 A4 G#4 F#4 E4 D#4 C#4

C#4 D#4 E4 F#4 G#4 A4 B4 C#5

Fingering coverage varies by selection; this atlas is a note reference.

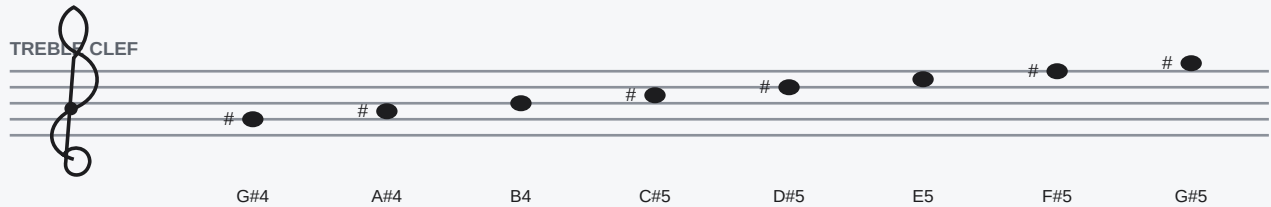
G# Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: G#4 - A#4 - B4 - C#5 - D#5 - E5 - F#5 - G#5

Fingers: Not source-checked; notes only

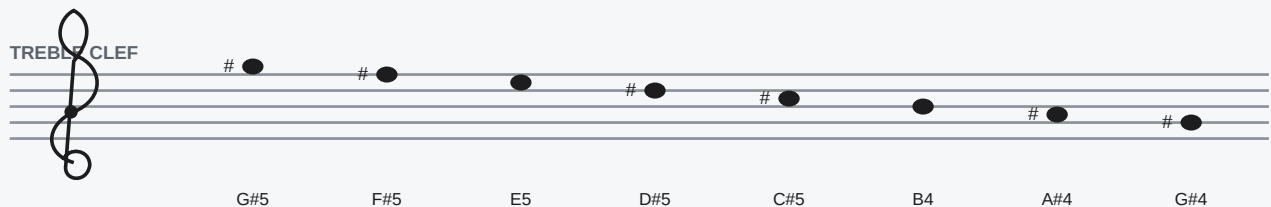


Descending · high to low

Actual ordered pitches

Notes: G#5 - F#5 - E5 - D#5 - C#5 - B4 - A#4 - G#4

Fingers: Not source-checked; notes only



Fingering coverage varies by selection; this atlas is a note reference.

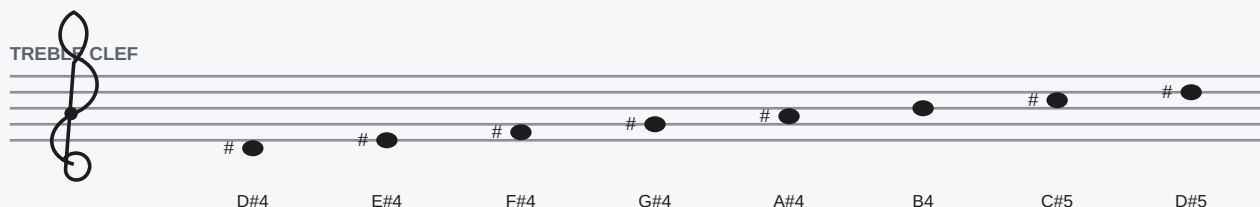
D# Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: D#4 - E#4 - F#4 - G#4 - A#4 - B4 - C#5 - D#5

Fingers: Not source-checked; notes only



Descending · high to low

Actual ordered pitches

Notes: D#5 - C#5 - B4 - A#4 - G#4 - F#4 - E#4 - D#4

Fingers: Not source-checked; notes only



Fingering coverage varies by selection; this atlas is a note reference.

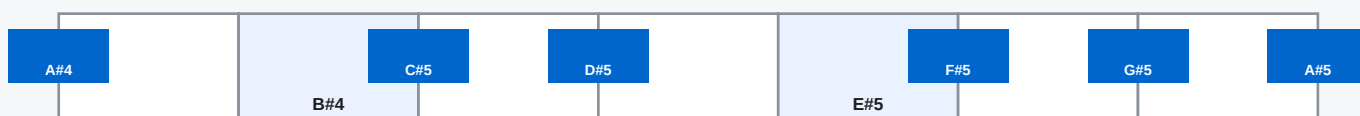
A# Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: A#4 - B#4 - C#5 - D#5 - E#5 - F#5 - G#5 - A#5

Fingers: Not source-checked; notes only

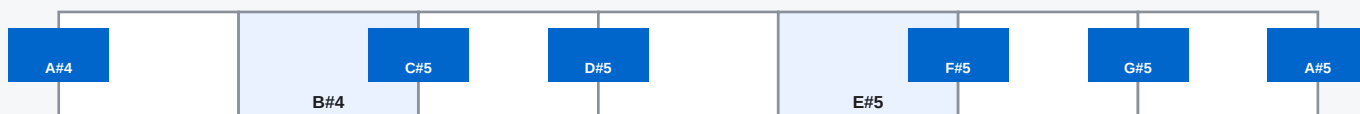


Descending · high to low

Actual ordered pitches

Notes: A#5 - G#5 - F#5 - E#5 - D#5 - C#5 - B#4 - A#4

Fingers: Not source-checked; notes only



Fingering coverage varies by selection; this atlas is a note reference.

D Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: D4 - E4 - F4 - G4 - A4 - Bb4 - C5 - D5

Fingers: Not source-checked; notes only

TREBLE CLEF



D4E4F4G4A4Bb4C5D5

D4	E4	F4	G4	A4		C5	D5
----	----	----	----	----	--	----	----

Descending · high to low

Actual ordered pitches

Notes: D5 - C5 - Bb4 - A4 - G4 - F4 - E4 - D4

Fingers: Not source-checked; notes only

TREBLE CLEF



D5C5Bb4A4G4F4E4D4

D4	E4	F4	G4	A4		C5	D5
----	----	----	----	----	--	----	----

Fingering coverage varies by selection; this atlas is a note reference.

G Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: G4 - A4 - Bb4 - C5 - D5 - Eb5 - F5 - G5

Fingers: Not source-checked; notes only

TREBLE CLEF

G4 A4 Bb4 C5 D5 Eb5 F5 G5

G4	A4	Bb4	C5	D5	Eb5	F5	G5
----	----	-----	----	----	-----	----	----

Descending · high to low

Actual ordered pitches

Notes: G5 - F5 - Eb5 - D5 - C5 - Bb4 - A4 - G4

Fingers: Not source-checked; notes only

TREBLE CLEF

G5 F5 Eb5 D5 C5 Bb4 A4 G4

G4	A4	Bb4	C5	D5	Eb5	F5	G5
----	----	-----	----	----	-----	----	----

Fingering coverage varies by selection; this atlas is a note reference.

C Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: C4 - D4 - Eb4 - F4 - G4 - Ab4 - Bb4 - C5

Fingers: Not source-checked; notes only

TREBLE CLEF

C4 D4 Eb4 F4 G4 Ab4 Bb4 C5

C4		Eb4		F4		Ab4	Bb4	C5
----	--	-----	--	----	--	-----	-----	----

Descending · high to low

Actual ordered pitches

Notes: C5 - Bb4 - Ab4 - G4 - F4 - Eb4 - D4 - C4

Fingers: Not source-checked; notes only

TREBLE CLEF

C5 Bb4 Ab4 G4 F4 Eb4 D4 C4

C4		Eb4		F4		Ab4	Bb4	C5
----	--	-----	--	----	--	-----	-----	----

Fingering coverage varies by selection; this atlas is a note reference.

F Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: F4 - G4 - Ab4 - Bb4 - C5 - Db5 - Eb5 - F5

Fingers: Not source-checked; notes only

TREBLE CLEF

F4 G4 Ab4 Bb4 C5 Db5 Eb5 F5

F4		G4	Ab4	Bb4		C5	Db5	Eb5		F5
----	--	----	-----	-----	--	----	-----	-----	--	----

Descending · high to low

Actual ordered pitches

Notes: F5 - Eb5 - Db5 - C5 - Bb4 - Ab4 - G4 - F4

Fingers: Not source-checked; notes only

TREBLE CLEF

F5 Eb5 Db5 C5 Bb4 Ab4 G4 F4

F4		G4	Ab4	Bb4		C5	Db5	Eb5		F5
----	--	----	-----	-----	--	----	-----	-----	--	----

Fingering coverage varies by selection; this atlas is a note reference.

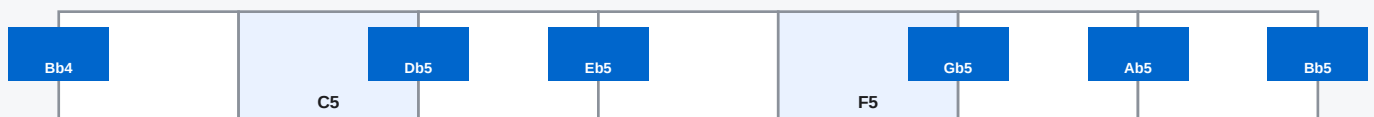
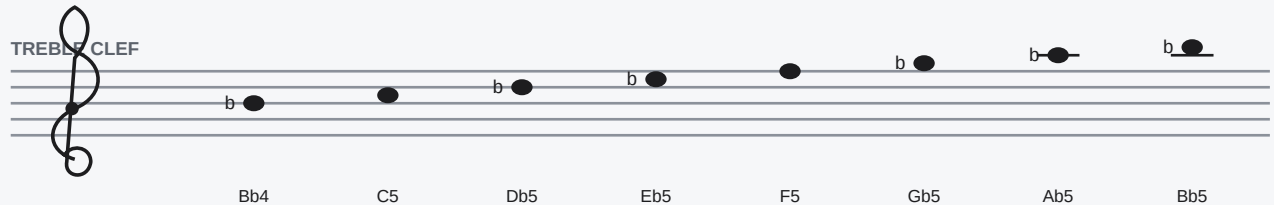
Bb Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: Bb4 - C5 - Db5 - Eb5 - F5 - Gb5 - Ab5 - Bb5

Fingers: Not source-checked; notes only

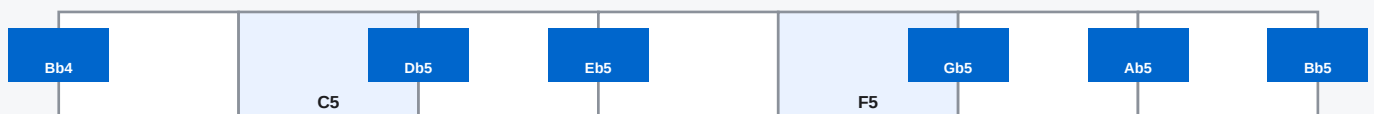
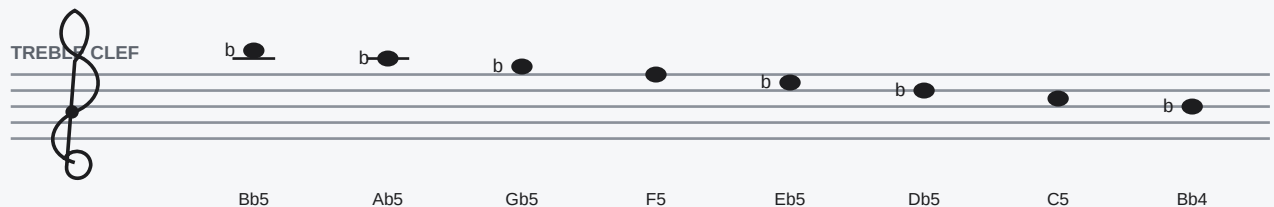


Descending · high to low

Actual ordered pitches

Notes: Bb5 - Ab5 - Gb5 - F5 - Eb5 - Db5 - C5 - Bb4

Fingers: Not source-checked; notes only



Fingering coverage varies by selection; this atlas is a note reference.

F# Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: Eb4 - F4 - Gb4 - Ab4 - Bb4 - Cb5 - Db5 - Eb5

Fingers: Not source-checked; notes only

TREBLE CLEF

Notes on staff: Eb4, F4, Gb4, Ab4, Bb4, Cb5, Db5, Eb5

Piano Roll Visualization:

Eb4		F4	Gb4	Ab4	Bb4	Cb5		Db5	Eb5
-----	--	----	-----	-----	-----	-----	--	-----	-----

Descending · high to low

Actual ordered pitches

Notes: Eb5 - Db5 - Cb5 - Bb4 - Ab4 - Gb4 - F4 - Eb4

Fingers: Not source-checked; notes only

TREBLE CLEF

Notes on staff: Eb5, Db5, Cb5, Bb4, Ab4, Gb4, F4, Eb4

Piano Roll Visualization:

Eb4		F4	Gb4	Ab4	Bb4	Cb5		Db5	Eb5
-----	--	----	-----	-----	-----	-----	--	-----	-----

Fingering coverage varies by selection; this atlas is a note reference.

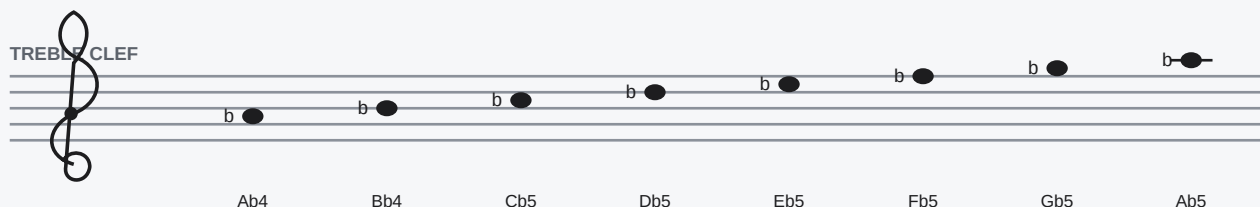
Ah Natural Minor

Ascending · low to high

Actual ordered pitches

Notes: Ab4 - Bb4 - Cb5 - Db5 - Eb5 - Fb5 - Gb5 - Ab5

Fingers: Not source-checked; notes only

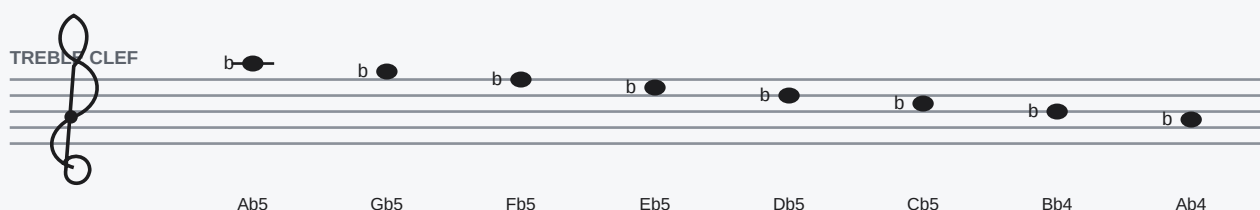


Descending · high to low

Actual ordered pitches

Notes: Ab5 - Gb5 - Fb5 - Eb5 - Db5 - Cb5 - Bb4 - Ab4

Fingers: Not source-checked; notes only



Fingering coverage varies by selection; this atlas is a note reference.

A Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: A4 - B4 - C5 - D5 - E5 - F5 - G#5 - A5

Fingers: Not source-checked; notes only

TREBLE CLEF

A4

B4

C5

D5

E5

F5

G#5

A5

A4		B4		C5		D5		E5		F5		G#5		A5
----	--	----	--	----	--	----	--	----	--	----	--	-----	--	----

Descending · high to low

Actual ordered pitches

Notes: A5 - G#5 - F5 - E5 - D5 - C5 - B4 - A4

Fingers: Not source-checked; notes only

TREBLE CLEF

A5

G#5

F5

E5

D5

C5

B4

A4

A4		B4		C5		D5		E5		F5		G#5		A5
----	--	----	--	----	--	----	--	----	--	----	--	-----	--	----

Fingering coverage varies by selection; this atlas is a note reference.

F Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: E4 - F#4 - G4 - A4 - B4 - C5 - D#5 - E5

Fingers: Not source-checked; notes only

Descending · high to low

Actual ordered pitches

Notes: E5 - D#5 - C5 - B4 - A4 - G4 - F#4 - E4

Fingers: Not source-checked; notes only

TREBLE CLEF

E5 D#5 C5 B4 A4 G4 F#4 E4

E4 G4 A4 B4 C5 D#5 E5

Fingering coverage varies by selection; this atlas is a note reference.

B Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: B4 - C#5 - D5 - E5 - F#5 - G5 - A#5 - B5

Fingers: Not source-checked; notes only

TREBLE CLEF

B4 C#5 D5 E5 F#5 G5 A#5 B5

B4	C#5	D5	E5	F#5	G5	A#5	B5
----	-----	----	----	-----	----	-----	----

Descending · high to low

Actual ordered pitches

Notes: B5 - A#5 - G5 - F#5 - E5 - D5 - C#5 - B4

Fingers: Not source-checked; notes only

TREBLE CLEF

B5 A#5 G5 F#5 E5 D5 C#5 B4

B4	C#5	D5	E5	F#5	G5	A#5	B5
----	-----	----	----	-----	----	-----	----

Fingering coverage varies by selection; this atlas is a note reference.

F# Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: F#4 - G#4 - A4 - B4 - C#5 - D5 - E#5 - F#5

Fingers: Not source-checked; notes only

TREBLE CLEF

F#4 G#4 A4 B4 C#5 D5 E#5 F#5

F#4	G#4			C#5			F#5
		A4	B4		D5		E#5

Descending · high to low

Actual ordered pitches

Notes: F#5 - E#5 - D5 - C#5 - B4 - A4 - G#4 - F#4

Fingers: Not source-checked; notes only

TREBLE CLEF

F#5 E#5 D5 C#5 B4 A4 G#4 F#4

F#4	G#4			C#5			F#5
		A4	B4		D5		E#5

Fingering coverage varies by selection; this atlas is a note reference.

C# Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: C#4 - D#4 - E4 - F#4 - G#4 - A4 - B#4 - C#5

Fingers: Not source-checked; notes only

TREBLE CLEF

C#4 D#4 E4 F#4 G#4 A4 B#4 C#5

C#4	D#4	E4	F#4	G#4	A4	B#4	C#5
-----	-----	----	-----	-----	----	-----	-----

Descending · high to low

Actual ordered pitches

Notes: C#5 - B#4 - A4 - G#4 - F#4 - E4 - D#4 - C#4

Fingers: Not source-checked; notes only

TREBLE CLEF

C#5 B#4 A4 G#4 F#4 E4 D#4 C#4

C#4	D#4	E4	F#4	G#4	A4	B#4	C#5
-----	-----	----	-----	-----	----	-----	-----

Fingering coverage varies by selection; this atlas is a note reference.

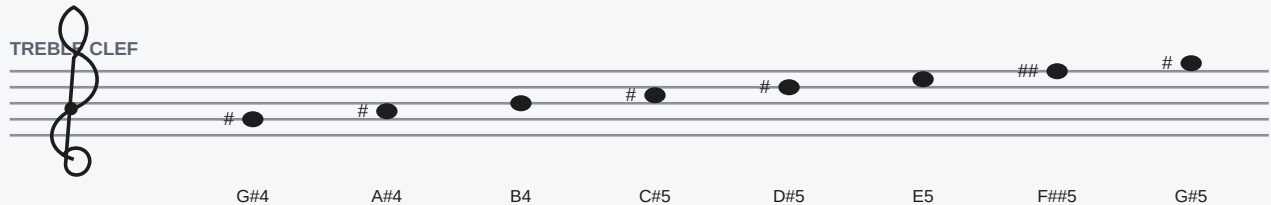
G# Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: G#4 - A#4 - B4 - C#5 - D#5 - E5 - F##5 - G#5

Fingers: Not source-checked; notes only

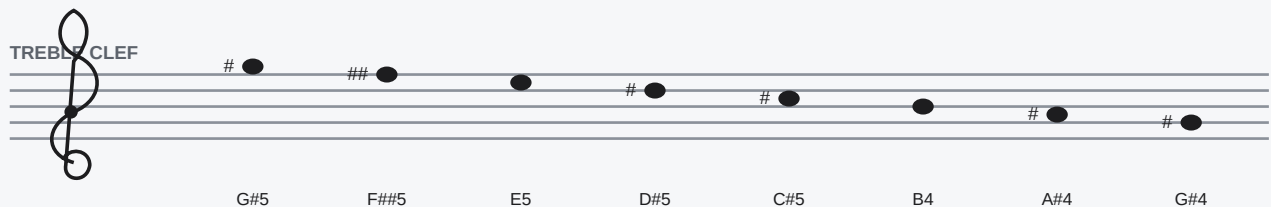


Descending · high to low

Actual ordered pitches

Notes: G#5 - F##5 - E5 - D#5 - C#5 - B4 - A#4 - G#4

Fingers: Not source-checked; notes only



Fingering coverage varies by selection; this atlas is a note reference.

D# Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: D#4 - E#4 - F#4 - G#4 - A#4 - B4 - C##5 - D#5

Fingers: Not source-checked; notes only

TREBLE CLEF

D#4 E#4 F#4 G#4 A#4 B4 C##5 D#5

D#4		E#4	F#4	G#4	A#4	B4		C##5	D#5
-----	--	-----	-----	-----	-----	----	--	------	-----

Descending · high to low

Actual ordered pitches

Notes: D#5 - C##5 - B4 - A#4 - G#4 - F#4 - E#4 - D#4

Fingers: Not source-checked; notes only

TREBLE CLEF

D#5 C##5 B4 A#4 G#4 F#4 E#4 D#4

D#4		E#4	F#4	G#4	A#4	B4		C##5	D#5
-----	--	-----	-----	-----	-----	----	--	------	-----

Fingering coverage varies by selection; this atlas is a note reference.

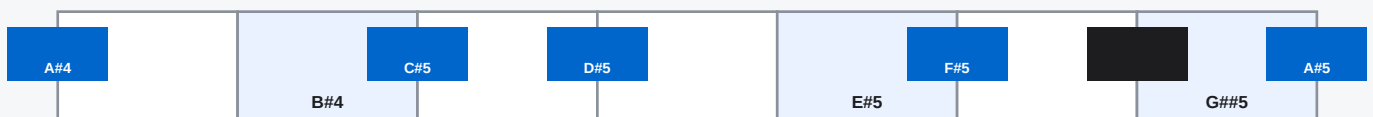
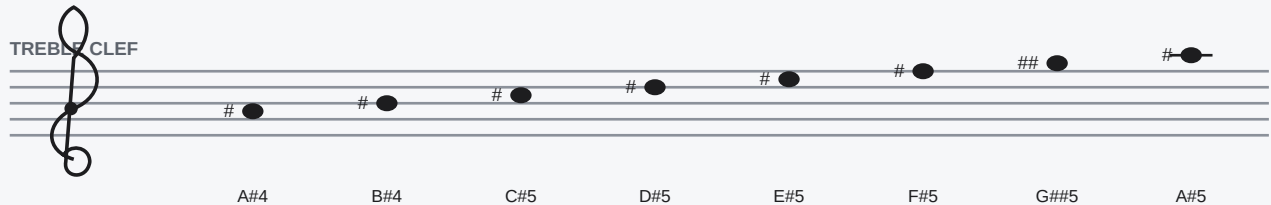
A# Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: A#4 - B#4 - C#5 - D#5 - E#5 - F#5 - G##5 - A#5

Fingers: Not source-checked; notes only

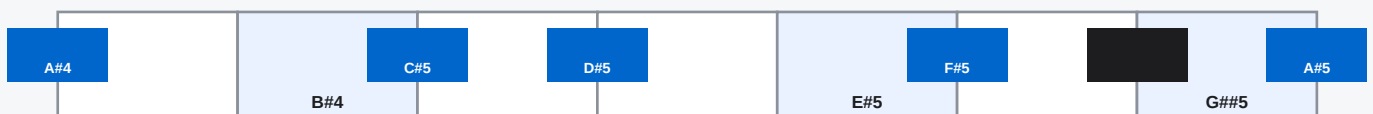


Descending · high to low

Actual ordered pitches

Notes: A#5 - G##5 - F#5 - E#5 - D#5 - C#5 - B#4 - A#4

Fingers: Not source-checked; notes only



Fingering coverage varies by selection; this atlas is a note reference.

D Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: D4 - E4 - F4 - G4 - A4 - Bb4 - C#5 - D5

Fingers: Not source-checked; notes only

TREBLE CLEF

D4E4F4G4A4Bb4C#5D5

D4	E4	F4	G4	A4			
				Bb4		C#5	D5

Descending · high to low

Actual ordered pitches

Notes: D5 - C#5 - Bb4 - A4 - G4 - F4 - E4 - D4

Fingers: Not source-checked; notes only

TREBLE CLEF

D5C#5Bb4A4G4F4E4D4

D4	E4	F4	G4	A4			
				Bb4		C#5	D5

Fingering coverage varies by selection; this atlas is a note reference.

G Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: G4 - A4 - Bb4 - C5 - D5 - Eb5 - F#5 - G5

Fingers: Not source-checked; notes only

TREBLE CLEF

G4 A4 Bb4 C5 D5 Eb5 F#5 G5

G4	A4	Bb4	C5	D5	Eb5	F#5	G5
----	----	-----	----	----	-----	-----	----

Descending · high to low

Actual ordered pitches

Notes: G5 - F#5 - Eb5 - D5 - C5 - Bb4 - A4 - G4

Fingers: Not source-checked; notes only

TREBLE CLEF

G5 F#5 Eb5 D5 C5 Bb4 A4 G4

G4	A4	Bb4	C5	D5	Eb5	F#5	G5
----	----	-----	----	----	-----	-----	----

Fingering coverage varies by selection; this atlas is a note reference.

C Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: C4 - D4 - Eb4 - F4 - G4 - Ab4 - B4 - C5

Fingers: Not source-checked; notes only

TREBLE CLEF

C4 D4 Eb4 F4 G4 Ab4 B4 C5

C4		Eb4		F4		Ab4		B4	C5
----	--	-----	--	----	--	-----	--	----	----

Descending · high to low

Actual ordered pitches

Notes: C5 - B4 - Ab4 - G4 - F4 - Eb4 - D4 - C4

Fingers: Not source-checked; notes only

TREBLE CLEF

C5 B4 Ab4 G4 F4 Eb4 D4 C4

C4		Eb4		F4		Ab4		B4	C5
----	--	-----	--	----	--	-----	--	----	----

Fingering coverage varies by selection; this atlas is a note reference.

F Harmonic Minor

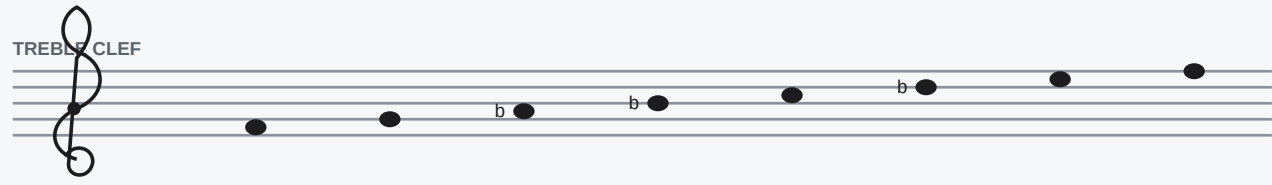
Ascending · low to high

Actual ordered pitches

Notes: F4 - G4 - Ab4 - Bb4 - C5 - Db5 - E5 - F5

Fingers: Not source-checked; notes only

TREBLE CLEF



F4

G4

Ab4

Bb4

C5

Db5

E5

F5

F4		Ab4	Bb4		Db5		E5	F5
----	--	-----	-----	--	-----	--	----	----

Descending · high to low

Actual ordered pitches

Notes: F5 - E5 - Db5 - C5 - Bb4 - Ab4 - G4 - F4

Fingers: Not source-checked; notes only

TREBLE CLEF



F5

E5

Db5

C5

Bb4

Ab4

G4

F4

F4		Ab4	Bb4		Db5		E5	F5
----	--	-----	-----	--	-----	--	----	----

Fingering coverage varies by selection; this atlas is a note reference.

Rh Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: Bb4 - C5 - Db5 - Eb5 - F5 - Gb5 - A5 - Bb5

Fingers: Not source-checked; notes only

TREBLE CLEF

A musical staff with a treble clef. The notes are placed on the lines and spaces from bottom to top: Bb4 (flat, space 1), C5 (line 1), Db5 (flat, space 2), Eb5 (flat, line 2), F5 (space 3), Gb5 (flat, line 3), A5 (space 4), and Bb5 (flat, line 4).

Bb4 C5 Db5 Eb5 F5 Gb5 A5 Bb5

The diagram shows a 12-tone scale with notes Bb4, C5, Db5, Eb5, F5, Gb5, A5, and Bb5. The notes are represented by colored boxes: Bb4 (blue), C5 (light blue), Db5 (blue), Eb5 (blue), F5 (light blue), Gb5 (blue), A5 (black), and Bb5 (blue). The notes are arranged in a sequence, with Bb4 and Bb5 being the same note an octave apart.

Descending · high to low

Actual ordered pitches

Notes: Bb5 - A5 - Gb5 - F5 - Eb5 - Db5 - C5 - Bb4

Fingers: Not source-checked; notes only

TREBLE CLEF

Bb5 A5 Gb5 F5 Eb5 Db5 C5 Bb4

The diagram shows a 12-tone scale with chromatic alterations. The notes are arranged in a row, with a line above them. The notes are: Bb4, C5, Db5, Eb5, F5, Gb5, A5, and Bb5. The notes C5, F5, and A5 are in light blue boxes, while the others are in dark blue boxes. The notes Bb5 and Gb5 are in black boxes. The notes Bb4, Db5, Eb5, F5, and A5 are labeled with their respective note names and octave numbers.

Fingering coverage varies by selection; this atlas is a note reference.

Fh Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: Eb4 - F4 - Gb4 - Ab4 - Bb4 - Cb5 - D5 - Eb5

Fingers: Not source-checked; notes only

TREBLE CLEF

A musical staff with a treble clef. The notes are placed on the staff lines and spaces as follows: Eb4 (flat, first space), F4 (first line), Gb4 (flat, second space), Ab4 (flat, second line), Bb4 (flat, third space), Cb5 (flat, third line), D5 (third space), and Eb5 (flat, fourth space).

Note	Staff Position
E _b 4	First Space
F4	First Line
G _b 4	Second Space
A _b 4	Second Line
B _b 4	Third Space
C _b 5	Third Line
D5	Third Space
E _b 5	Fourth Space

[illegible]

Descending · high to low

Actual ordered pitches

Notes: Eb5 - D5 - Cb5 - Bb4 - Ab4 - Gb4 - F4 - Eb4

Fingers: Not source-checked; notes only

TREBLE CLEF

Notes on the staff: Eb5, D5, Cb5, Bb4, Ab4, Gb4, F4, Eb4.

Diagram of a 5-qubit quantum circuit. The circuit has five qubits. Qubit 4 (index 4) is initialized to $|1\rangle$. The circuit consists of several gates: a CNOT from qubit 4 to qubit 0, a CNOT from qubit 4 to qubit 1, a CNOT from qubit 4 to qubit 2, a CNOT from qubit 4 to qubit 3, and a CNOT from qubit 4 to qubit 5. The final state of the circuit is a 5-qubit state with qubit 4 being the only one in state $|1\rangle$.

Fingering coverage varies by selection; this atlas is a note reference.

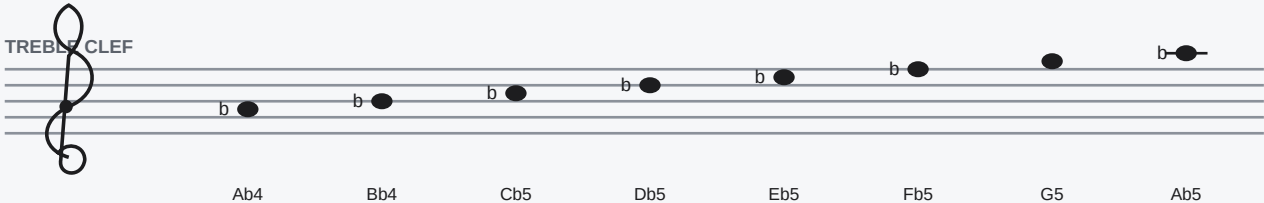
Ah Harmonic Minor

Ascending · low to high

Actual ordered pitches

Notes: Ab4 - Bb4 - Cb5 - Db5 - Eb5 - Fb5 - G5 - Ab5

Fingers: Not source-checked; notes only

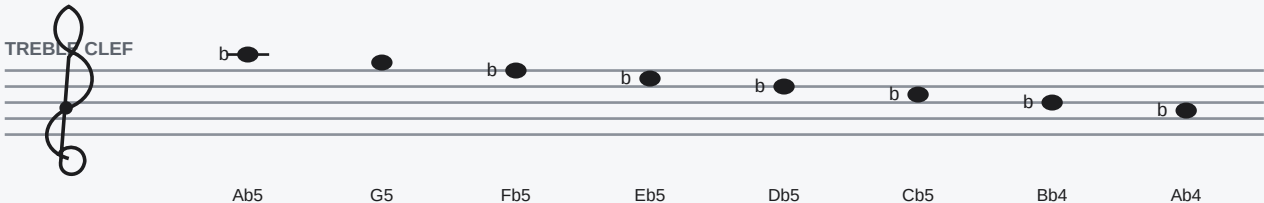


Descending · high to low

Actual ordered pitches

Notes: Ab5 - G5 - Fb5 - Eb5 - Db5 - Cb5 - Bb4 - Ab4

Fingers: Not source-checked; notes only



Fingering coverage varies by selection; this atlas is a note reference.

A Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: A4 - B4 - C5 - D5 - E5 - F#5 - G#5 - A5

Fingers: Not source-checked; notes only

TREBLE CLEF

A4 B4 C5 D5 E5 F#5 G#5 A5

A4	B4	C5	D5	E5	F#5	G#5	A5
----	----	----	----	----	-----	-----	----

Descending · high to low

Actual ordered pitches

Notes: A5 - G5 - F5 - E5 - D5 - C5 - B4 - A4

Fingers: Not source-checked; notes only

TREBLE CLEF

A5 G5 F5 E5 D5 C5 B4 A4

A4	B4	C5	D5	E5	F5	G5	A5
----	----	----	----	----	----	----	----

Classical melodic minor prints its distinct ascent and natural-minor descent.

F Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: E4 - F#4 - G4 - A4 - B4 - C#5 - D#5 - E5

Fingers: Not source-checked; notes only

TREBLE CLEF

E4 F#4 G4 A4 B4 C#5 D#5 E5

E4		F#4					C#5		D#5		E5
----	--	-----	--	--	--	--	-----	--	-----	--	----

Descending · high to low

Actual ordered pitches

Notes: E5 - D5 - C5 - B4 - A4 - G4 - F#4 - E4

Fingers: Not source-checked; notes only

TREBLE CLEF

E5 D5 C5 B4 A4 G4 F#4 E4

E4		F#4					C5		D5		E5
----	--	-----	--	--	--	--	----	--	----	--	----

Classical melodic minor prints its distinct ascent and natural-minor descent.

B Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: B4 - C#5 - D5 - E5 - F#5 - G#5 - A#5 - B5

Fingers: Not source-checked; notes only

TREBLE CLEF

B4 C#5 D5 E5 F#5 G#5 A#5 B5

B4	C#5	D5	E5	F#5	G#5	A#5	B5
----	-----	----	----	-----	-----	-----	----

Descending · high to low

Actual ordered pitches

Notes: B5 - A5 - G5 - F#5 - E5 - D5 - C#5 - B4

Fingers: Not source-checked; notes only

TREBLE CLEF

B5 A5 G5 F#5 E5 D5 C#5 B4

B4	C#5	D5	E5	F#5	G5	A5	B5
----	-----	----	----	-----	----	----	----

Classical melodic minor prints its distinct ascent and natural-minor descent.

F# Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: F#4 - G#4 - A4 - B4 - C#5 - D#5 - E#5 - F#5

Fingers: Not source-checked; notes only

TREBLE CLEF

F#4G#4A4B4C#5D#5E#5F#5

F#4	G#4			C#5	D#5		E#5	F#5
		A4	B4					

Descending · high to low

Actual ordered pitches

Notes: F#5 - E5 - D5 - C#5 - B4 - A4 - G#4 - F#4

Fingers: Not source-checked; notes only

TREBLE CLEF

F#5E5D5C#5B4A4G#4F#4

F#4	G#4			C#5				F#5
		A4	B4		D5	E5		

Classical melodic minor prints its distinct ascent and natural-minor descent.

C# Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: C#4 - D#4 - E4 - F#4 - G#4 - A#4 - B#4 - C#5

Fingers: Not source-checked; notes only

TREBLE CLEF

C#4 D#4 E4 F#4 G#4 A#4 B#4 C#5

C#4	D#4		F#4	G#4	A#4		C#5
	E4					B#4	

Descending · high to low

Actual ordered pitches

Notes: C#5 - B4 - A4 - G#4 - F#4 - E4 - D#4 - C#4

Fingers: Not source-checked; notes only

TREBLE CLEF

C#5 B4 A4 G#4 F#4 E4 D#4 C#4

C#4	D#4		F#4	G#4			C#5
	E4			A4	B4		

Classical melodic minor prints its distinct ascent and natural-minor descent.

G# Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: G#4 - A#4 - B4 - C#5 - D#5 - E#5 - F##5 - G#5

Fingers: Not source-checked; notes only

TREBLE CLEF

G#4 A#4 B4 C#5 D#5 E#5 F##5 G#5

G#4	A#4		C#5	D#5		E#5		F##5	G#5
		B4							

Descending · high to low

Actual ordered pitches

Notes: G#5 - F#5 - E5 - D#5 - C#5 - B4 - A#4 - G#4

Fingers: Not source-checked; notes only

TREBLE CLEF

G#5 F#5 E5 D#5 C#5 B4 A#4 G#4

G#4	A#4		C#5	D#5		F#5		G#5
		B4			E5			

Classical melodic minor prints its distinct ascent and natural-minor descent.

D# Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: D#4 - E#4 - F#4 - G#4 - A#4 - B#4 - C##5 - D#5

Fingers: Not source-checked; notes only

TREBLE CLEF

D#4		E#4	F#4	G#4	A#4		B#4	C##5	D#5
-----	--	-----	-----	-----	-----	--	-----	------	-----

Descending · high to low

Actual ordered pitches

Notes: D#5 - C#5 - B4 - A#4 - G#4 - F#4 - E#4 - D#4

Fingers: Not source-checked; notes only

TREBLE CLEF

D#4		E#4	F#4	G#4	A#4	B4		C#5	D#5
-----	--	-----	-----	-----	-----	----	--	-----	-----

Classical melodic minor prints its distinct ascent and natural-minor descent.

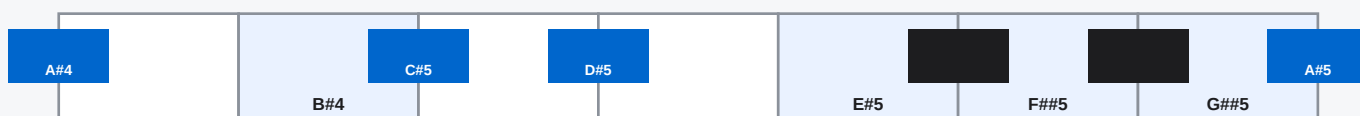
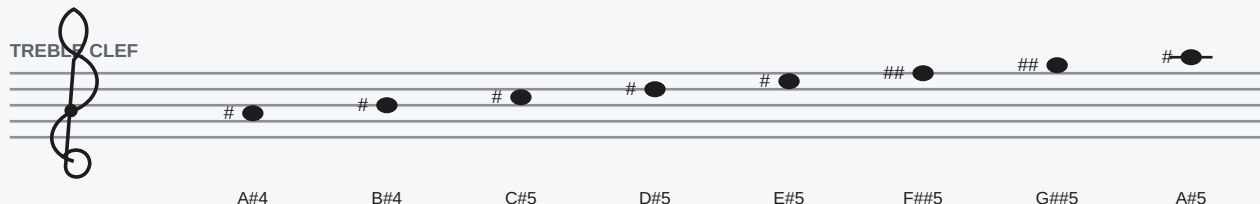
A# Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: A#4 - B#4 - C#5 - D#5 - E#5 - F##5 - G##5 - A#5

Fingers: Not source-checked; notes only

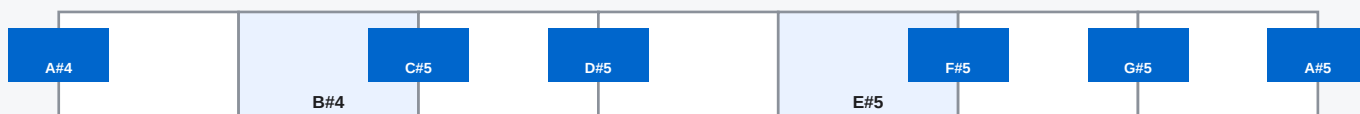


Descending · high to low

Actual ordered pitches

Notes: A#5 - G#5 - F#5 - E#5 - D#5 - C#5 - B#4 - A#4

Fingers: Not source-checked; notes only



Classical melodic minor prints its distinct ascent and natural-minor descent.

D Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: D4 - E4 - F4 - G4 - A4 - B4 - C#5 - D5

Fingers: Not source-checked; notes only

TREBLE CLEF

D4 E4 F4 G4 A4 B4 C#5 D5

D4	E4	F4	G4	A4	B4	C#5	D5
----	----	----	----	----	----	-----	----

Descending · high to low

Actual ordered pitches

Notes: D5 - C5 - Bb4 - A4 - G4 - F4 - E4 - D4

Fingers: Not source-checked; notes only

TREBLE CLEF

D5 C5 Bb4 A4 G4 F4 E4 D4

D4	E4	F4	G4	A4	Bb4	C5	D5
----	----	----	----	----	-----	----	----

Classical melodic minor prints its distinct ascent and natural-minor descent.

G Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: G4 - A4 - Bb4 - C5 - D5 - E5 - F#5 - G5

Fingers: Not source-checked; notes only

TREBLE CLEF

G4 A4 Bb4 C5 D5 E5 F#5 G5

G4	A4	Bb4	C5	D5	E5	F#5	G5
----	----	-----	----	----	----	-----	----

Descending · high to low

Actual ordered pitches

Notes: G5 - F5 - Eb5 - D5 - C5 - Bb4 - A4 - G4

Fingers: Not source-checked; notes only

TREBLE CLEF

G5 F5 Eb5 D5 C5 Bb4 A4 G4

G4	A4	Bb4	C5	D5	Eb5	F5	G5
----	----	-----	----	----	-----	----	----

Classical melodic minor prints its distinct ascent and natural-minor descent.

C Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: C4 - D4 - Eb4 - F4 - G4 - A4 - B4 - C5

Fingers: Not source-checked; notes only

TREBLE CLEF

C4 D4 Eb4 F4 G4 A4 B4 C5

C4	D4		F4	G4	A4	B4	C5
----	----	--	----	----	----	----	----

Descending · high to low

Actual ordered pitches

Notes: C5 - Bb4 - Ab4 - G4 - F4 - Eb4 - D4 - C4

Fingers: Not source-checked; notes only

TREBLE CLEF

C5 Bb4 Ab4 G4 F4 Eb4 D4 C4

C4	D4		F4	G4			C5
----	----	--	----	----	--	--	----

Classical melodic minor prints its distinct ascent and natural-minor descent.

F Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: F4 - G4 - Ab4 - Bb4 - C5 - D5 - E5 - F5

Fingers: Not source-checked; notes only

TREBLE CLEF

F4	G4	Ab4	Bb4	C5	D5	E5	F5
----	----	-----	-----	----	----	----	----

Descending · high to low

Actual ordered pitches

Notes: F5 - Eb5 - Db5 - C5 - Bb4 - Ab4 - G4 - F4

Fingers: Not source-checked; notes only

TREBLE CLEF

F4	G4	Ab4	Bb4	C5	Db5	Eb5	F5
----	----	-----	-----	----	-----	-----	----

Classical melodic minor prints its distinct ascent and natural-minor descent.

Bb Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: Bb4 - C5 - Db5 - Eb5 - F5 - G5 - A5 - Bb5

Fingers: Not source-checked; notes only

TREBLE CLEF

Bb4 C5 Db5 Eb5 F5 G5 A5 Bb5

Bb4		Db5	Eb5		F5		G5		A5		Bb5
	C5										

Descending · high to low

Actual ordered pitches

Notes: Bb5 - Ab5 - Gb5 - F5 - Eb5 - Db5 - C5 - Bb4

Fingers: Not source-checked; notes only

TREBLE CLEF

Bb5 Ab5 Gb5 F5 Eb5 Db5 C5 Bb4

Bb4		Db5	Eb5		F5		Gb5		Ab5		Bb5
	C5										

Classical melodic minor prints its distinct ascent and natural-minor descent.

F# Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: Eb4 - F4 - Gb4 - Ab4 - Bb4 - C5 - D5 - Eb5

Fingers: Not source-checked; notes only

TREBLE CLEF

Diagram illustrating the ascending scale (low to high) in Eb4. The notes are: Eb4, F4, Gb4, Ab4, Bb4, C5, D5, and Eb5. The notes are displayed on a musical staff with a Treble Clef. Below the staff, the notes are mapped to a piano grid layout.

Eb4		F4	Gb4	Ab4	Bb4		C5	D5	Eb5
-----	--	----	-----	-----	-----	--	----	----	-----

Descending · high to low

Actual ordered pitches

Notes: Eb5 - Db5 - Cb5 - Bb4 - Ab4 - Gb4 - F4 - Eb4

Fingers: Not source-checked; notes only

TREBLE CLEF

Diagram illustrating the descending scale (high to low) in Eb4. The notes are: Eb5, Db5, Cb5, Bb4, Ab4, Gb4, F4, and Eb4. The notes are displayed on a musical staff with a Treble Clef. Below the staff, the notes are mapped to a piano grid layout.

Eb4		F4	Gb4	Ab4	Bb4	Cb5		Db5	Eb5
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Classical melodic minor prints its distinct ascent and natural-minor descent.

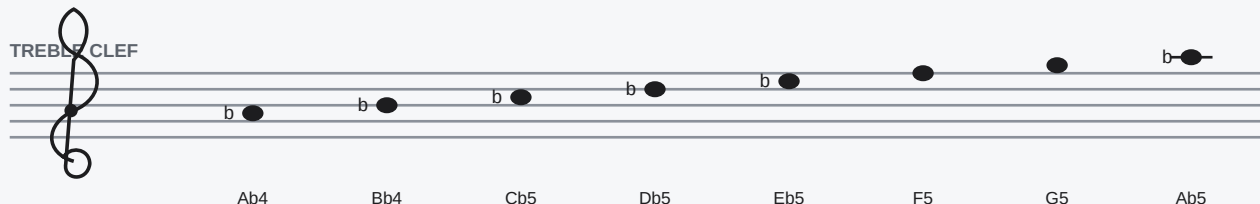
Ab Classical Melodic Minor

Ascending · low to high

Actual ordered pitches

Notes: Ab4 - Bb4 - Cb5 - Db5 - Eb5 - F5 - G5 - Ab5

Fingers: Not source-checked; notes only

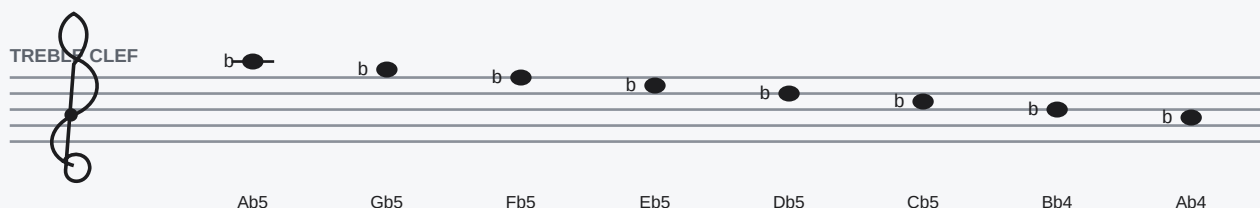


Descending · high to low

Actual ordered pitches

Notes: Ab5 - Gb5 - Fb5 - Eb5 - Db5 - Cb5 - Bb4 - Ab4

Fingers: Not source-checked; notes only



Classical melodic minor prints its distinct ascent and natural-minor descent.